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CHAPTER ONE BASIC DEFINITIONS

Thermodynamics is the study of heat related to matter in its ability for energy

1.0 Thermodynamics is the study of energy, its transformation, and effects on states of matter. The word that would probably first come to your mind as you start this course is temperature, this is so because the sensation of heat which, is subjectively expressed as either hot or cold is an everyday phenomenon in the course of human daily activities. The traditional definition of temperature is that it is a measure of the degree of hotness or coldness of a body and result from energy transfer when a temperature difference occurs between two bodies in thermal contact. Temperature is commonly measured on the Celsius temperature scale between two fixed points, which corresponds to the freezing (0°C) and boiling points (100°C) of water.

0°C freezing point of water
100°C boiling point of water

In engineering however, temperature is measured on the absolute thermodynamic temperature scale which is independent of the material the thermometer is made from and hence does not vary from material to material. On this scale, the absolute zero is defined as the fraction $1/273.15$ of the thermodynamic temperature of the triple point of water (0.01°C). At absolute zero the molecules of a gas are stationary, would occupy zero volume and exert zero pressure. It is the absolute limit of temperature. The absolute temperature scale has unit in Kelvin (K) in honour of the British physicist Lord Kelvin who did some ground breaking pioneering work in the field of low temperature research. The Celsius temperature scale can be converted to absolute scale by adding 273.

For example,

$$25^{\circ}\text{C} = 25 + 273 = 298 \text{ K}$$

✓ A heat source is a body having a higher heat capacity than another body with which it is in thermal contact and heat transfer can take place between the two bodies. Examples are the sun and the oven.

✓ A heat sink is a body that is capable of receiving heat from another body that has a higher heat capacity and with which it is in thermal contact. Examples include the atmosphere, the sea and a deep freezer.

Thermodynamics is the study of heat related to matter in its

Kelvin-plant formula
Clausius formula

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27. Are thermal conductors also good electrical conductors at normal and high temperatures?
28. Define pressure and state its unit of measurement.
29. State the origin of atmospheric temperature.
30. What height of a column of mercury can the atmospheric pressure support at 0°C . State the relationship between gauge pressure and absolute pressure.
31. Distinguish between gauge pressure, absolute pressure and atmospheric pressure.
32. What is the weight of a 70 kg astronaut on the surface of the moon?
33. Differentiate between mass, density, specific volume and specific gravity.
34. Describe what a vacuum is and state its unit of measurement.
35. Describe a device for vacuum measurement.
36. Describe with the aid of sketches, two methods of measuring pressure.
37. What height of column of mercury can atmospheric pressure support?
38. Explain the principle behind the use of piezometer or pressure tube to measure the intensity of pressure in a fluid.
39. State the atmospheric pressure in the following units: Pascal, N/m^2 , KN/mm^2 , mmHg, mmH₂O, bar.
40. With the aid of a sketch, explain the principle of operation of the bourdon pressure gauge.
41. State the density of fresh water, sea water, air, petrol and mercury.
42. State the average human blood pressure in mm of mercury at full breath and between breaths. What pumps the blood in a human being and explain how it works?
43. What blood pressure level is usually considered dangerous for an adult human?
44. Where on the body would you measure the blood pressure?
45. State and describe, with the aid of sketches, the principle of operation of an instrument for measuring the human blood pressure.

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47
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Basic definitions

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A heat reservoir is a body having a large heat capacity, part of which can be transferred to another body to keep it at a constant temperature. For example, a cake in an oven.

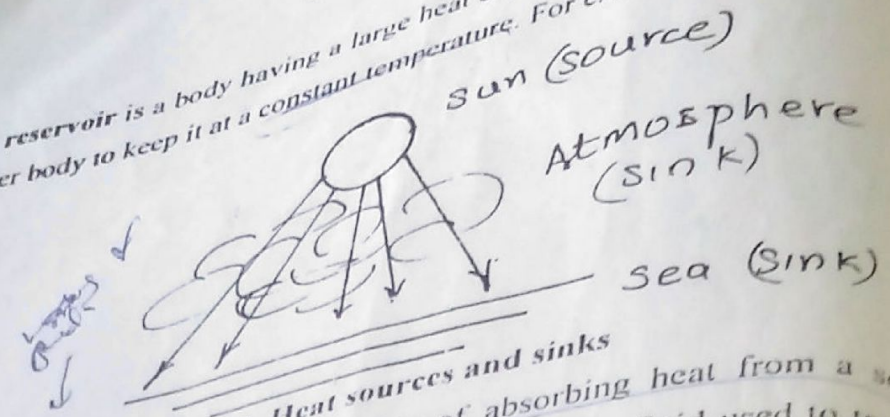


Fig. 1.1 Heat sources and sinks

Working fluid is a substance that is capable of absorbing heat from a source and rejecting it to a heat sink. For example, water is the working fluid used to transfer heat from the motor car engine to the air with the aid of a radiator.

Fuel is a substance having a large amount of chemical energy that can be converted into work and light forms of energy by a combustion process. Examples are kerosene, wood, cooking gas, petrol and diesel.

Heat engine is a system that converts heat energy to mechanical energy. Heat engine can be either internal or external combustion. An example of an internal combustion heat engine is a motor car engine that converts the heat energy in the petrol to mechanical energy for driving the wheels of the car.

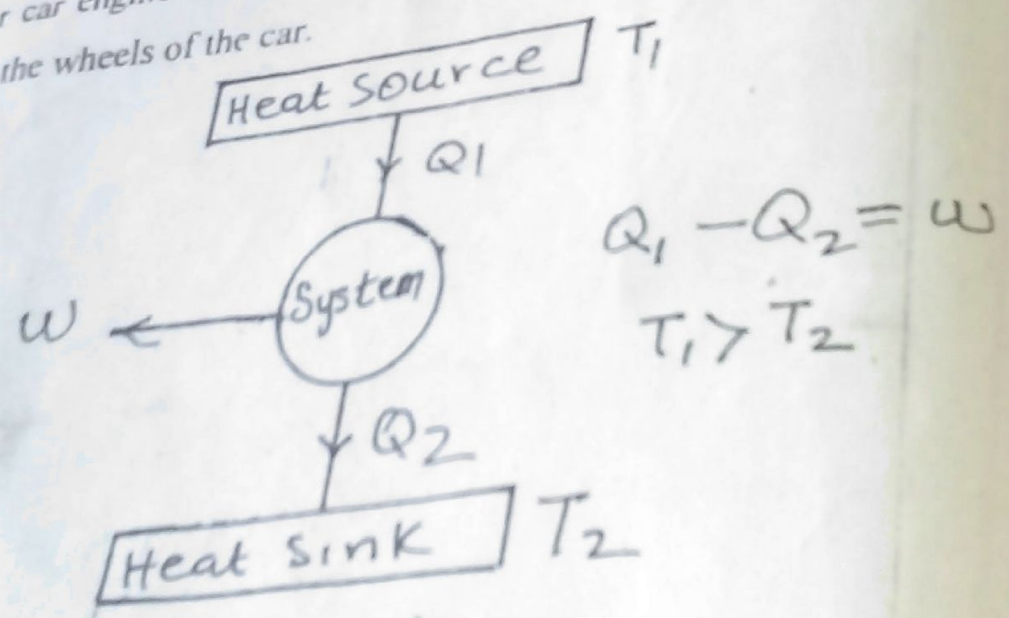


Fig. 1.2 Heat engine.

Heat pump is a system that absorbs heat from an external low temperature source and rejects it completely without converting any part of it to work, to an external sink that is at a higher temperature. It is the reverse of heat engine and a typical example is the domestic refrigerator where heat is absorbed from items stored in the cooling compartment and rejected to the air around the coil of tube behind the refrigerator. To achieve this it requires a work input to initiate this effect and a compressor is commonly used for this.

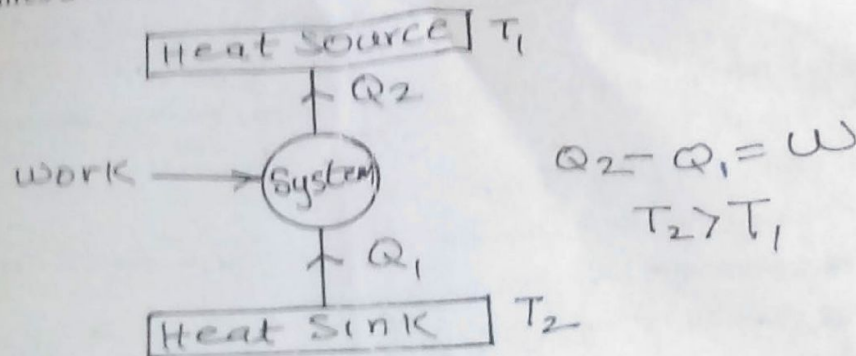


Fig. 1.3 Heat pump.

1.2. Problems

1. Define thermodynamic
2. Define temperature and state its unit of measurement
3. Define the absolute temperature scale and state its unit of measurement.
4. State the relationship between the normal and the absolute temperature scales.
5. Define the state of the molecules of a real gas at the absolute zero temperature.
6. What is the main benefit of the absolute temperature scale?
7. State the average ambient temperature in Nigeria, Alaska, Saudi Arabia, Russian Siberia and England.
8. State the temperature of the earth's crust.
9. State the temperature of the sun, venues, Jupiter, mercury and the dark and lit surfaces of the moon.

10. State the normal human blood temperature and the threshold at which it becomes dangerous.
11. State the freezing temperature of fresh water and saline (sea) water.
12. State ten properties of a substance that varies with temperature.
13. If you add salt to ice, would you expect the melting point to increase or decrease? Explain.
14. State the freezing and boiling points of water, alcohol, ammonia, oxygen and carbon dioxide.
15. Why is rain water colder than borehole water?
16. Can increased pressure raise the boiling point of water? Explain and state typical domestic and industrial applications.
17. State the melting point of the following substances: lead, copper, steel, diamond, burnt brick and mercury.
18. Draw a clinical thermometer, explain the function of the various parts and indicate the temperature range.
19. Define the Kelvin scale of temperature.
20. Why is water not a suitable liquid for a thermometer?
21. State three places on the human body where a thermometer can be placed to measure the body temperature.
22. Describe the principle of an experiment to check the "ice" and "steam" points of a thermometer.
23. Sketch and explain the principle of operation, stating the equation (if any) relating the properties of the following types of thermometer: thermocouple, bimetal strip, platinum resistance, optical pyrometer and gas.
24. Explain how respiration evaporation can reduce the body temperature.
25. Describe the changes in colour that occurs when steel is heated to a very high temperature.
26. Distinguish between thermal and electrical conductivity at atomic level and state in each case the unit of measurement.

27. Are thermal conductors also good electrical conductors at normal and high temperatures?
28. Define pressure and state its unit of measurement.
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Basic definitions

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Control of work when
force is used through
distance

46. Define heat. Is heat a property or an interaction? Explain.
47. Define heat source, heat sink, heat reservoir. State two examples of each.
48. Distinguish between a working fluid and a fuel. State five examples of each.
49. Describe the heat engine and explain the difference between internal and external combustion engines. State an example of each.
50. Describe the heat pump and state one example and explain how it works.
51. Define the critical point of a substance undergoing a phase change and state the critical pressure and temperature of water.
52. Define saturation.
53. Describe how the chemical-change and physical-change of a substance can be used for temperature measurement.

Any substance is a fluid capable of deformation either by compression or expansion and in the process of transferring energy from place to another.

A substance is a single substance or mixture of substances have the same composition throughout.

It is an actual or hypothetical envelope surrounding the the condition of the system at an instant of time as measured by its properties.

A quantity of matter with an homogeneous chemical and physical structure.

A phenomenon of change from one state to another a quantity of matter isolated within a boundary where under investigation.

CHAPTER TWO FUNDAMENTAL CONCEPTS IN THERMODYNAMICS

Types of system
Open system
Closed system
Isolated system

Definition of Basic Terms:

System is all the matter enclosed within a specified space whose characteristics can be specified and measured. A system can occur in two forms, **flow** and **non-flow**. A flow or open system allows matter and energy to flow in and/or out of the system. A non-flow or closed system contains a fixed quantity of matter. No matter is allowed to flow in or out, but thermal, electrical and mechanical forms of energy may flow in or out.

An isolated system is a system that has no interaction with its surroundings.

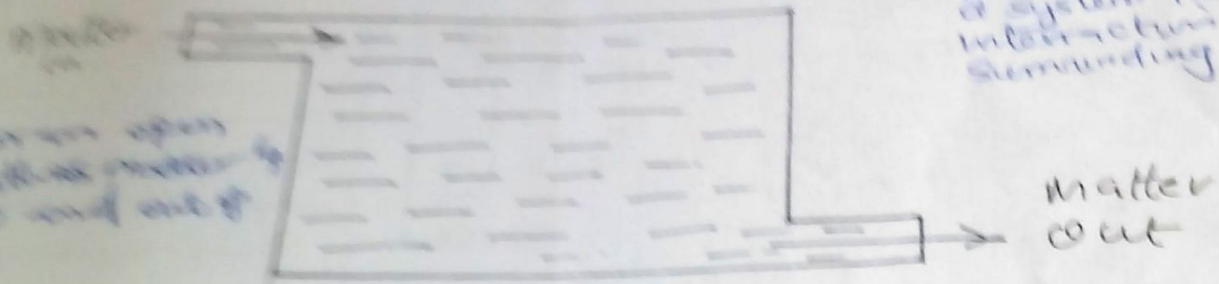
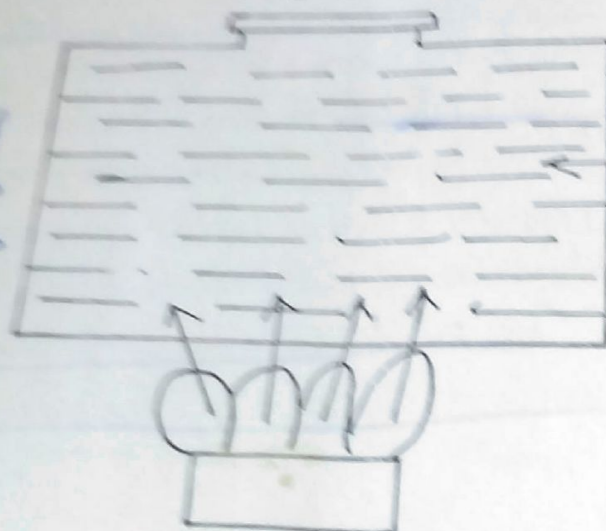


Fig. 2.1 Flow system



Heat Source

Extensive properties directly proportional to quantity of matter in system

Intensive properties independent of quantity of matter in the system.

Fig. 2.2 Non-flow system

Work is a quantity of matter isolated within a boundary and is under investigation. It is the transient (moving) quantity of energy being described which a force is moved through a distance. Ex: 80 m/s

Boundary is a line which may be real or imaginary that defines the limit of a system. A boundary can be drawn to follow the outline of the system or enclose it in a required form. It is fixed if the system itself is fixed or moving if the system moves.

Surrounding is all the matter that lies immediately outside the boundary and interacts with it. Most thermodynamic systems are usually in thermal or mechanical contact with the surrounding which may be air, water or oil.

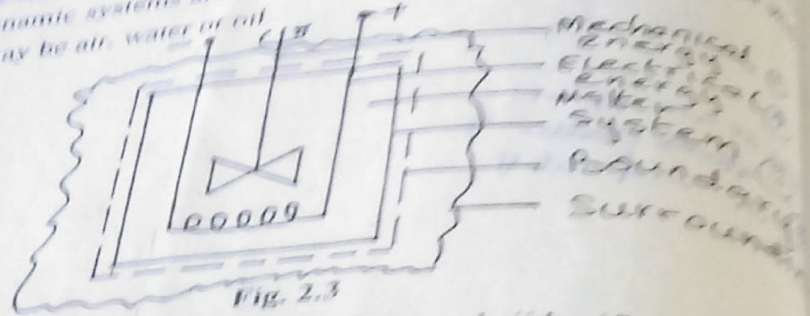


Fig. 2.3

Property is any observable characteristics of a system that can be identified, measured and varied in any required manner. The fundamental properties are pressure, volume and temperature. A property is a state function because it depends on the initial and final states of the system and not the type of change. Two types of properties can be identified in a system, the extensive property which is directly proportional to the quantity of matter in the system and has discrete units of measurement e.g. temperature, volume m^3 and mass kg and the intensive property that is independent of the quantity of matter in the system and the units are specified as per extensive property e.g. pressure N/m^2 and density kg/m^3 .

The State of a system at any given instant is its condition, which is fully defined by specifying its properties. (At a given state, if two independent properties are known, the state can be fully defined by using the equation of state that relates the properties of the system. This is known as the two-property rule and enables unknown properties to be determined.)

property is the quantity of matter in the system and has units of measurement
property is independent of the quantity of matter in the system

Process is the change that takes place when the properties of a system are varied. When the change does not occur at an instant the process is fully defined by stating its initial and final properties, the path (if identifiable) followed and the interactions that take place across system boundaries during the process. A process can be flow or non-flow and can be represented on a diagram on which the axes are the properties of the process. For example, the expansion of a gas in a cylinder in which the pressure and volume varies as shown in Fig. 2.4. Points 1, 2, 3, 4 are states while line 1 - 4, is the path of the process.

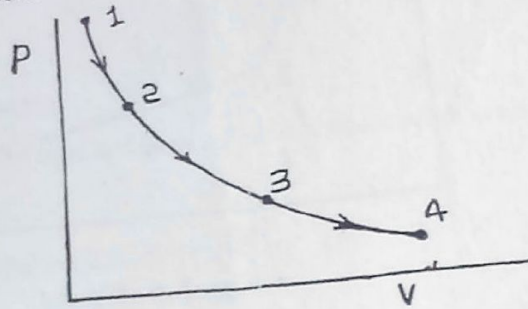


Fig. 2.4. Process

Cycle is a process or sequence of processes in which the initial and the final states of the system are the same. A cycle diagram can be drawn by combining the diagrams for all the processes. It should be numbered sequentially so that the first process is 1 - 2, followed by 2 - 3 and so on until it ends at 4-1 to close the cycle.

In Fig. 2.5, the cycle is 1 - 2 - 3 - 4 - 1

A thermodynamic system is a region in space or quantity of matter bounded by a closed surface. The surroundings include everything external to the system and the involve system is separated from the surroundings by the system boundaries. These boundaries can be movable or fixed, real or imaginary. The interactions between thermal systems and their environment are described via work, heat transfer, and material flow streams.

bounded

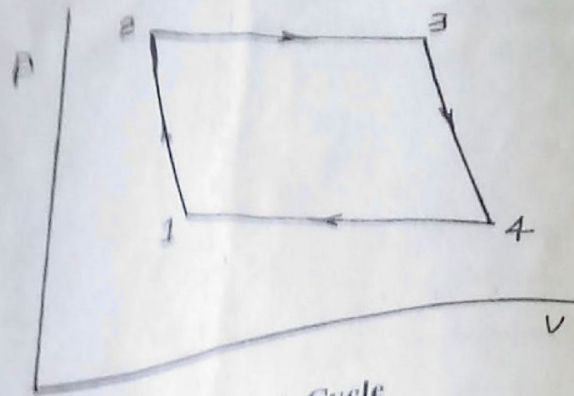


Fig. 2.5 Cycle

2.1. Equation of State

- ✓ The equation of state of a pure substance is a mathematical relationship between the pressure, specific volume and temperature of the substance. When the system is in thermodynamic equilibrium the relationship between the properties is,

$$f(p, v, T) = 0$$

The equation of state enables on an unknown property to be determined when two are known.

2.2. Kinetic Theory of Gases

Most thermodynamic changes are based on the kinetic theory of gases, which assumed that gases consist of an infinitely large amount of molecules moving at very high velocity and in an undefined and randomized manner. In the process they collide with each other and the wall of the container.

No net change in kinetic energy occurs when the molecules collide because the gases are ideal and do not experience intermolecular attraction. The mean kinetic energy of each molecule is directly proportional to the absolute temperature of the gas. The volumes of the molecules themselves are very small so that the mean distance between them is large compared with their sizes.

The velocities in the x, y, z axes are u, v, w respectively as shown in Fig. 2.6.

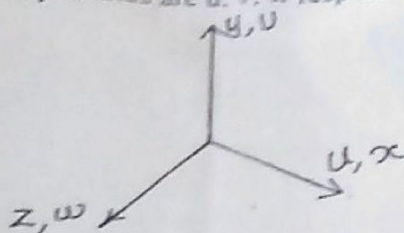


Fig. 2.6 Axis system for velocity

The pressure p is due to the average effects of molecules with a range of velocities. To account for this we introduce

$$\begin{aligned}\text{The root mean square molecular velocity} &= \frac{1}{3}(u^2 + v^2 + w^2) \\ &= \frac{1}{3}c^2\end{aligned}$$

If the volume of the gas is v and n molecules are contained with each having a mass m , then the pressure resulting from the collision

$$p = \frac{n}{3v} mc^2$$

Thus

$$pv = \frac{1}{3} nmc^2 \quad (2.1)$$

Since the average kinetic energy is directly proportional to the absolute temperature of the gas, i.e.

$$\frac{1}{2} mc^2 \propto T$$

Then

$$mc^2 = 2kT$$

Which if substituted in equation (2.1) gives

$$pv = \frac{2}{3} nkT$$

Avogadro's law

According to Avogadro's hypothesis, equal volume of all gases under the same conditions of pressure and temperature contains the same number of molecules.